

Marcus, 38, ~85 kg · pulled his daughter from a house fire · mixed-depth burn · circumferential forearms, anterior trunk, partial face · tachycardic, hypotensive

BIO 004 · WEEK 1 · CASE DEEP DIVE

The Integumentary System

A burn case that asks the skin to tell you what's left, what's gone, and what it will take to keep him alive

PATIENT CASE

Marcus is 38 years old, approximately 85 kilograms. About forty minutes ago he ran back into a **house fire** to pull his seven-year-old daughter out. She is in the adjacent bay, awake. He is not.

His face has **soot around the nostrils** and patchy redness across the forehead and cheeks. The **anterior surface of his chest and abdomen** is a deep angry red, weeping, and **exquisitely tender** when the trauma nurse passes a gloved finger near it. Both **forearms and the dorsum of both hands** tell a different story: **pale, waxy white** in some patches, **charred black** in others, **leathery to the touch**, and to his apparent surprise he reports **no pain at all** when those areas are examined. His **fingertips beyond the burns are cool and faintly dusky**.

HR 132 · **BP** 96/58 · **RR** 28 · **SpO₂** 94% on room air · **Temp** 36.4°C

The senior nurse needs **three numbers** in the next few minutes: **how much of his body surface is burned, how much fluid he will need in the next twenty-four hours, and which of these burned areas she should be most worried about right now.**

GUIDING QUESTION

Which skin layers were destroyed where, what does that destruction take with it, and how do you turn a body covered in two different kinds of burn into a fluid prescription and a triage plan?

PREWORK

This is the case-based companion. The terms, the video, and the spaced recall for the integumentary system live on your Week 1 prework page. Open both side by side: the prework gives you the inventory; this page gives you the burn that uses every term.

1 THE LARGEST ORGAN YOU FORGET IS AN ORGAN

Skin does more than wrap you

The integument is the body's largest organ. It covers about two square meters of an adult and weighs more than the liver. We tend to think of it as packaging, but when a chunk of it is destroyed the consequences are immediate and systemic. Marcus is in shock not from a wound to a deep organ but from the loss of his outer shell.

Hold onto five jobs the skin is doing right now: it is a **barrier** against pathogens and fluid loss, a **thermoregulator** through sweat and surface vasoconstriction, a **sensory organ** packed with nerve endings, a **metabolic surface** that makes vitamin D, and an **immunologic interface** studded with Langerhans cells. Every one of those jobs ends abruptly where the burn destroyed the skin that did them.

Why he is in shock. Burns destroy the barrier. Plasma weeps directly into the interstitium and the air. A 40% burn loses several liters per hour into nothing. The vascular space empties, the heart speeds up to compensate, and the blood pressure drops anyway. The fluid he needs is not extra. It is replacement for what is leaking out of him through skin that is no longer there.

Epidermis, dermis, hypodermis

Skin is built in three layers. Each one has a job, and each one determines what is at stake when a burn reaches that depth.

EPIDERMIS

The outermost layer, made of stratified squamous keratinized epithelium. No blood vessels of its own; it gets nutrients by diffusion from the dermis below. It is constantly being replaced from a single layer of dividing cells at its base, the *stratum basale*. Above the basale, cells migrate upward through the *stratum spinosum*, fill with keratin in the *stratum granulosum*, pass through a thin *stratum lucidum* only in thick skin (palms, soles), and finally die and flatten into the *stratum corneum* that you actually see and touch. The whole cycle takes about a month. Pigment-producing *melanocytes* sit in the basale and donate melanin to the keratinocytes above them. Immune-sentinel *Langerhans cells* live in the spinosum.

Why this matters for Marcus: a burn that destroys only the epidermis (a first-degree burn, like a sunburn) leaves the basale intact. The basal cells re-epithelialize the area in days, no scar. A burn that destroys the basale and reaches the dermis cannot re-epithelialize from the surface anymore; the cells that replace skin are gone.

DERMIS

Below the epidermis sits the dermis, a thick layer of connective tissue with two zones. The *papillary dermis* is loose, vascular, and projects upward into the epidermis as papillae that anchor the two layers together. The *reticular dermis* is denser, made of irregular collagen and elastin, and it carries the blood vessels, lymphatics, sensory receptors, hair follicles, sweat glands, and sebaceous glands.

Almost everything you intuitively think of as "what skin does" lives in the dermis. The sensory nerve endings that report pain and touch sit here. So do the capillary loops that flush red with embarrassment, blanch pale with fear, and leak when a partial-thickness burn raises a blister. A burn that reaches deep into the reticular dermis takes those structures with it.

HYPODERMIS (SUBCUTANEOUS TISSUE, SUPERFICIAL FASCIA)

Below the dermis but technically not part of the skin: a layer of areolar connective tissue and adipose. It cushions, insulates, and anchors the skin to the underlying fascia and muscle. A burn that destroys all of the dermis and continues into the hypodermis is called full-thickness

(third-degree). Past the hypodermis into muscle, tendon, or bone makes it a fourth-degree burn.

Hair, glands, nails, and the nerves that report back

Embedded in the dermis are structures that originated as folds of epidermis pushed downward during development. They retain the capacity to grow epidermal cells, which is why **partial-thickness burns can heal from below**: even if the surface epidermis is gone, hair follicles and sweat glands deep in the dermis still contain basal-type cells that can climb back up and reform the epidermal surface. Destroy the dermis entirely and you destroy these reservoirs. Surface healing then has to come from the wound edges, which is much slower and produces scar.

Hair follicles

Tubes of epidermal cells extending into the dermis. Each houses a hair shaft, a sebaceous gland, and a tiny smooth muscle (arrector pili) that produces goosebumps.

Sebaceous (oil) glands

Open into hair follicles; produce sebum, an oily lubricant that waterproofs the hair and skin and has mild antimicrobial properties.

Ecocrine sweat glands

Distributed over almost the entire body. Coiled tubes deep in the dermis. Their watery secretion cools the body by evaporation; thermoregulation depends on them.

Apocrine sweat glands

Localized to axillae, areolae, perineum. Open into hair follicles. Secretion is thicker, becomes odorous when bacteria act on it.

Nails

Hardened plates of keratin on the dorsum of digits. Grow continuously from the nail matrix under the proximal fold.

Sensory receptors

Meissner corpuscles (light touch, papillary dermis), Pacinian corpuscles (deep pressure and vibration, reticular dermis/hypodermis), Ruffini endings (stretch), Merkel discs (sustained touch), and free nerve endings (pain and temperature).

This is why Marcus's full-thickness areas are **painless**. The free nerve endings that report pain live in the dermis. When the dermis is destroyed, the nociceptors go with it. The injury is the worst kind he has, and he cannot feel it. Painful areas on his chest tell you the dermis is partially intact there. The waxy white painless areas on his forearms tell you the dermis is gone.

First through fourth degree

Burn depth is named by the deepest skin layer destroyed. Each depth has a characteristic appearance, pain pattern, and healing trajectory.

DEPTH	LAYER DESTROYED	APPEARANCE	PAIN	HEALING
First (superficial)	epidermis only	red, dry, no blisters	painful	days, no scar (sunburn)
Second, superficial partial-thickness	epidermis + papillary dermis	red, moist, blisters, blanches with pressure	severely painful	2-3 weeks, minimal scar
Second, deep partial-thickness	epidermis + into reticular dermis	pink-white, may not blanch, drier	less painful (some nociceptors destroyed)	3-9 weeks, scar likely; may need grafting
Third (full-thickness)	entire epidermis + entire dermis +/- hypodermis	waxy white, leathery, black eschar, dry	painless (nociceptors destroyed)	cannot re-epithelialize; requires grafting; contracts
Fourth	through hypodermis into muscle, tendon, or bone	charred, exposed deep structures	painless	surgical reconstruction, possible amputation

Marcus has at least three depths visible at once. The blistered painful chest and abdomen are *superficial partial-thickness*. The waxy white, painless, leathery forearm areas are *full-thickness*. The patches of black eschar over the dorsum of his hands suggest *third or fourth degree*, possibly involving extensor tendons. Each region needs a different prediction about healing and a different surgical plan.

The rule of nines

You cannot prescribe fluid resuscitation without knowing how much of the body's surface area (TBSA) is burned. The rule of nines is a quick adult estimate that divides the body into regions, each one approximately a multiple of nine percent.

Head and neck

9% total (about 4.5% face, 4.5% back of head)

Each upper limb

9% (anterior 4.5%, posterior 4.5%); both arms = 18%

Anterior trunk

18% (chest 9%, abdomen 9%)

Posterior trunk

18% (upper back 9%, lower back 9%)

Each lower limb

18% (anterior 9%, posterior 9%); both legs = 36%

Perineum

1%

Two important caveats. First, the rule of nines is for *adults*; pediatric patients have proportionally larger heads, so the Lund and Browder chart is used instead. Second, only *partial-thickness and full-thickness* burns (second degree and deeper) count toward TBSA for fluid resuscitation. Superficial first-degree burns like sunburn are not included.

WORKED EXAMPLE, MARCUS

Both forearms and hands + anterior chest + anterior abdomen + partial face

Both arms circumferential, mid-forearm to fingertips, roughly half of each upper limb $\approx 9\%$ ($4.5\% \times 2$)

Anterior chest $\approx 9\%$

Anterior abdomen $\approx 9\%$

Anterior dorsal hands already counted in arms

Face involved patchily $\approx 4.5\%$, count half $\approx \sim 2\text{-}4\%$

Estimated TBSA $\approx 40\%$ (round to ease bedside math)

The Parkland formula

Once you have a TBSA estimate and a weight, you can prescribe fluid. The Parkland formula gives a starting point for the first twenty-four hours.

PARKLAND FORMULA

$4 \text{ mL} \times \text{weight in kg} \times \% \text{TBSA} = \text{total mL of Lactated Ringer's in the first 24 hours}$

Half of that total goes in over the **first 8 hours**, timed from the moment of the burn, not from when the patient arrived in the ED.

The remaining half goes in over the **next 16 hours**.

MARCUS'S CALCULATION

$4 \times 85 \text{ kg} \times 40\% = 13,600 \text{ mL Lactated Ringer's over 24 hours}$

First 8 hours from time of burn: $6,800 \text{ mL} = 850 \text{ mL/hour}$

The burn was ~40 minutes ago. The first 8-hour window has about 7 hours and 20 minutes left. The hourly rate is calculated against the time remaining in that window, not against a clean 8 hours.

Next 16 hours: $6,800 \text{ mL} = 425 \text{ mL/hour}$

Parkland is a starting estimate, not a prescription. Titrate to *urine output*: target 0.5 mL/kg/hour in adults, ~30-50 mL/hour for Marcus. Under-resuscitation causes shock and acute kidney injury; over-resuscitation causes pulmonary edema, abdominal compartment syndrome, and worsening tissue edema in the burn itself. The number on the page gets you started. The patient tells you what is actually needed.

What the skin is telling you across the room

The skin is the first physical-exam finding you see when you walk up to a patient. Color changes localize quickly because the dermis is so vascular and so close to the surface. Train yourself to read the four below; together they will catch most of what skin can tell you.

● **Erythema (red)**

Increased blood flow through dilated dermal capillaries. Inflammation, infection, fever, flushing, superficial burn. Blanches with pressure if vessels are intact.

● **Pallor (pale)**

Reduced perfusion. Vasoconstriction (cold, anxiety, shock), anemia, or destruction of dermal vessels (full-thickness burn appears waxy white because the dermal vasculature is gone).

● **Cyanosis (blue-gray)**

Deoxygenated hemoglobin visible through the skin. Central cyanosis (lips, oral mucosa) suggests hypoxemia. Peripheral cyanosis (fingertips, toes) suggests low cardiac output, vasoconstriction, or local circulatory obstruction.

● **Jaundice (yellow)**

Bilirubin accumulating in skin and sclera. Hepatocellular injury, biliary obstruction, hemolysis. In burn patients, watch for it later as a sign of hepatic dysfunction, sepsis, or massive hemolysis.

Marcus shows three of these on first look. **Erythema** on the partial-thickness chest, **pallor** on the full-thickness forearms, and **cyanosis** at his fingertips. The fingertip cyanosis is the most urgent finding on this list. He has circumferential burns around both forearms. As the tissue edema rises under the rigid, leathery eschar, the dermal and subcutaneous swelling has nowhere to expand outward. Pressure builds inward and compresses the distal vasculature. Cool, dusky fingertips on a circumferential burn is the sign that perfusion to the hand is being cut off from inside. The bedside intervention is an *escharotomy*: a longitudinal incision through the eschar to release the pressure and restore distal flow.

Four phases, and why depth changes the path

Every wound passes through the same four phases, overlapping in time. Depth determines which phase dominates and how cleanly the skin returns.

1. Hemostasis

Minutes. Platelets aggregate; the clotting cascade lays down a fibrin plug. Provides a temporary scaffold and stops blood loss.

2. Inflammation

Hours to days. Neutrophils arrive first and clear debris and bacteria; macrophages follow and signal the next phase. Redness, swelling, warmth, pain.

3. Proliferation

Days to weeks. Fibroblasts lay down collagen, capillaries sprout (angiogenesis), and granulation tissue fills the defect. Re-epithelialization climbs across the surface from intact basal cells at the wound edges and from accessory structures.

4. Remodeling

Weeks to years. Collagen is reorganized from disordered type III to stronger type I; the wound flattens, pales, and gains tensile strength (never more than ~80% of original).

What depth does to this sequence:

- **Superficial burns** finish proliferation within days because the basal cells of the epidermis are intact and re-epithelialization is fast across a thin surface defect.
- **Superficial partial-thickness burns** heal in two to three weeks because hair follicles and sweat glands in the upper dermis serve as islands of basal-type cells; the new epithelium climbs outward from each one and fuses.
- **Deep partial-thickness burns** heal slowly (three to nine weeks) and often scar because most accessory structures are destroyed; only a few remain to seed re-epithelialization.
- **Full-thickness burns** cannot re-epithelialize from below; there are no surviving basal cells. The wound either contracts and scars or, more commonly, requires *skin grafting* to close the defect with autologous epidermis and dermis from an unburned donor site.

Why scars contract. In a full-thickness wound that is allowed to heal without grafting, fibroblasts called *myofibroblasts* pull the wound edges together to close it. Over weeks the

contracted scar shortens and may cross joints, restricting motion. This is why early grafting of full-thickness burns over hands, faces, and joints is so important.



DRAW ZONE 1 -- Sketch the three layers of skin and label the accessory structures. Annotate where 1st, 2nd, 3rd, 4th degree burns stop.

Fine for the basal cell layer; bold for burn-depth arrows.



DRAW ZONE 2 -- Anterior body diagram. Write the rule-of-nines percentage beside each region. Shade the burned areas and total your TBSA plus Parkland volume.

Different colors for partial vs full thickness. Star the circumferential areas.

The integument told you everything you needed

You walked in to a body covered in two different kinds of burn and a set of vitals that told you he was in shock. You leave with a number for his TBSA, a number for his first-eight-hours fluid, a triage plan, and a prediction for which areas will heal and which will need surgery.

"What % of his body is burned?"

~40% TBSA (partial-thickness anterior trunk 18% + bilateral arms 18% + partial face ~4%). Count second-degree and deeper only.

"How much fluid in 24 hours?"

Parkland: $4 \text{ mL} \times 85 \text{ kg} \times 40\% = \mathbf{13,600 \text{ mL Lactated Ringer's}}$. First 6,800 mL in the first 8 hours from time of burn; remainder over the next 16 hours.

"Why no pain on his forearms?"

Full-thickness burn destroyed the dermis, including the free nerve endings (nociceptors) that report pain. Painless burn = bad burn.

"Why dusky fingertips?"

Circumferential forearm burn + rising tissue edema under rigid eschar = inward pressure that compresses distal vasculature. This is an **escharotomy** indication, not just shock.

"What's the prediction for each area?"

Anterior trunk (superficial partial-thickness): heal in 2-3 weeks, minimal scar. Forearms and hands (full-thickness): cannot re-epithelialize, need grafting. Face: depth dependent; cosmetic and functional priority.

"What colors am I watching for next?"

Central cyanosis (inhalation injury, hypoxemia). Jaundice over the next days (hepatic dysfunction from rhabdo, sepsis, or massive transfusion). Spreading erythema around burn edges (cellulitis, burn wound infection).

You did not need a deep knowledge of any one of these in isolation. You needed the integument as a system. **Three layers, a few accessory structures, four colors, four healing phases.**

Marcus's case touches every one of them, and the structure of the skin tells you exactly what is wrong with him and exactly what to do next.

When you go back to the prework page, the epidermal strata and the accessory structures will land differently. Each one is now attached to a patient who is alive because someone read his skin correctly.